

DESIGN AND IMPLEMENTATION OF A TRAFFIC LIGHT CONTROLLER USING VERILOG HDL

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1. Abstract

This report presents the comprehensive design and simulation of a Traffic Light Controller using Verilog Hardware Description Language (HDL). A traffic light controller is a fundamental sequential digital circuit used in modern transportation infrastructure to regulate the flow of vehicles at road intersections. The controller described in this report manages two sets of traffic lights — one for the main street and one for the cross street — cycling through green, yellow, and red phases in a coordinated and timed sequence.

The design is written in synthesizable Verilog code and follows standard finite state machine (FSM) design principles using a clocked sequential logic approach. A dedicated Verilog testbench is used to drive input stimuli and verify the correctness of each phase transition. Simulation is performed to generate waveforms that visually confirm functional correctness across multiple full cycles, including mid-cycle reset behavior. This report includes the full HDL source code, a code walkthrough, RTL schematic analysis, and simulation results. The design is suitable for FPGA implementation and academic study in digital systems engineering.

2. Introduction

A Traffic Light Controller is one of the most practical and widely recognized applications of digital sequential logic design. It is a clocked finite state machine (FSM) that accepts a clock input and an asynchronous reset, and cycles through a predefined sequence of states to control the lighting outputs at a road intersection. Each state corresponds to a specific phase of traffic signal operation, such as the main street showing green while the cross street shows red.

In modern transportation systems, traffic light controllers are implemented using programmable logic devices such as FPGAs and microcontrollers. Understanding their design is fundamental to digital electronics and embedded systems education. The controller must ensure that conflicting green signals never occur simultaneously — a critical safety requirement — and that each phase maintains its programmed duration accurately.

In this project, Verilog HDL is used to model a two-street traffic light controller at the Register Transfer Level (RTL). The design uses three separate always blocks: one for the counter that tracks elapsed time, one for FSM state transitions, and one for output (light) logic. The 3-bit output encoding represents the three light colors: bit 0 = Green, bit 1 = Yellow, bit 2 = Red, giving single-hot encoded output values of 3'b001 (Green), 3'b010 (Yellow), and 3'b100 (Red).

The design is verified using a testbench that applies a clock, triggers a reset, and runs the simulation long enough to observe multiple complete 32-second cycles, including a mid-cycle reset test to verify robustness.

3. Objectives

The primary objectives of this project are as follows:

- To design a synthesizable and functionally correct Traffic Light Controller module using Verilog HDL that adheres to standard sequential digital design practices.
- To implement a four-state Finite State Machine (FSM) with clearly defined state transitions driven by a free-running counter, ensuring accurate phase timing.
- To encode the traffic light outputs using a 3-bit one-hot representation for the green, yellow, and red signals of both the main street and cross street.
- To implement a 5-bit counter (light_count) that tracks elapsed time within each 32-second cycle, providing the timing reference for all state transitions.
- To ensure mutual exclusivity of green signals — both streets must never simultaneously show green, guaranteeing intersection safety.
- To implement an asynchronous reset that immediately returns the FSM to its initial state (main green, cross red) regardless of the current state or counter value.
- To write a comprehensive testbench in Verilog that applies clock stimuli, exercises the reset function, and runs the simulation for multiple complete cycles.
- To simulate the design using EDA tools (Vivado and ModelSim) and analyze the resulting waveforms to confirm proper state sequencing and timing.
- To produce an RTL schematic after synthesis to visualize the underlying hardware structure including the FSM state registers, counter logic, and output multiplexers.
- To evaluate the design's suitability for FPGA deployment by analyzing its sequential logic structure and resource utilization.

4. Design Methodology

The Traffic Light Controller is designed using sequential logic principles in Verilog HDL. The overall design methodology follows a top-down approach: the behavioral FSM specification is first written in Verilog, then simulated to verify timing and correctness, and finally synthesized to produce an RTL schematic.

4.1 Module Architecture

The controller is encapsulated in a single Verilog module named `traffic_controller`. It takes two inputs: a clock signal (`clk`) and an asynchronous reset (`rst`). It produces three outputs: a 3-bit register `main_st` representing the main street light state, a 3-bit register `cross_st` representing the cross street light state, and a 5-bit register `light_count` that exposes the internal timing counter for testbench observation. The 5-bit counter supports values 0 through 31, covering the complete 32-second cycle.

4.2 Finite State Machine (FSM) Design

A 2-bit state register (`state_reg`) encodes four FSM states using a 2-bit parameter encoding. The four states and their meanings are:

Opcode	State	Description
00	g_to_r (Main Green)	Main street GREEN (15s), Cross street

		RED
01	y_to_r (Main Yellow)	Main street YELLOW (3s), Cross street RED
10	r_to_g (Cross Green)	Main street RED, Cross street GREEN (10s)
11	r_to_y (Cross Yellow)	Main street RED, Cross street YELLOW (3s)

4.3 Timing Counter Logic

A dedicated always block sensitive to the positive clock edge and asynchronous reset drives the 5-bit light_count counter. On reset, the counter clears to zero. On each rising clock edge it increments by 1. When it reaches 31 (the last count of a 32-step cycle), it wraps back to 0, creating a continuous free-running cycle of 32 clock periods. State transitions are triggered when the counter reaches specific threshold values.

4.4 State Transition Logic

A second always block implements the FSM state register, also sensitive to the positive clock edge and asynchronous reset. On reset, the FSM returns to g_to_r (the initial state). Otherwise, on each clock edge, the case statement checks the current state and transitions to the next state if the counter has reached the appropriate threshold value. The following table shows the complete timing sequence:

Phase	Duration	light_count Range	Next State
Main Green (g_to_r)	15 seconds	0 → 14	y_to_r
Main Yellow (y_to_r)	3 seconds	15 → 17	r_to_g
Cross Green (r_to_g)	10 seconds	18 → 27	r_to_y
Cross Yellow (r_to_y)	3 seconds	28 → 30	g_to_r

4.5 Output Logic

A third always block drives the main_st and cross_st output registers on each positive clock edge based on the current FSM state. The 3-bit output encoding is: 3'b001 = Green (bit 0), 3'b010 = Yellow (bit 1), 3'b100 = Red (bit 2). The two streets are always in complementary states — when the main street is green or yellow, the cross street is red, and vice versa. This guarantees that both streets are never simultaneously green, satisfying the fundamental safety requirement.

5. Program Code & Dynamic Explanation

5.1 Verilog Design Code

The following is the complete Verilog HDL source code for the Traffic Light Controller module:

```
`timescale 1ns / 1ps
module traffic_controller (
    input clk,
    input rst,
    output reg [2:0] main_st,    // Main street traffic light state
    output reg [2:0] cross_st,  // Cross street traffic light state
    output reg [4:0] light_count // Counter for traffic light timing
);
    // State encoding
    parameter g_to_r = 2'b00,
               y_to_r = 2'b01,
               r_to_g = 2'b10,
               r_to_y = 2'b11;

    reg [1:0] state_reg;

    initial begin
        light_count = 0;
    end

    // Light count timing logic
    always @ (posedge clk or posedge rst) begin
        if (rst)
            light_count = 4'b0000;
        else if (light_count == 31)
            light_count <= 0;
        else
            light_count <= light_count + 1;
    end

    // State transition logic
    always @ (posedge clk or posedge rst) begin
        if (rst)
            state_reg <= g_to_r;
        else
            case (state_reg)
                g_to_r: if (light_count == 15) state_reg <= y_to_r;
                y_to_r: if (light_count == 18) state_reg <= r_to_g;
                r_to_g: if (light_count == 28) state_reg <= r_to_y;
                r_to_y: if (light_count == 31) state_reg <= g_to_r;
            endcase
    end

    // Traffic light state logic
    always @ (posedge clk) begin
        case (state_reg)
            g_to_r: begin
                main_st <= 3'b001; // Main street green
                cross_st <= 3'b100; // Cross street red
            end
            y_to_r: begin
                main_st = 3'b010; // Main street yellow
                cross_st = 3'b100; // Cross street red
            end
        endcase
    end
endmodule
```

```
        end
        r_to_g: begin
            main_st <= 3'b100; // Main street red
            cross_st <= 3'b001; // Cross street green
        end
        r_to_y: begin
            main_st <= 3'b100; // Main street red
            cross_st = 3'b010; // Cross street yellow
        end
    endcase
end
endmodule
```

5.2 Code Component Breakdown and Analysis

Each section of the Verilog code plays a specific role in the Traffic Light Controller design. Below is a detailed explanation of every component:

Module Declaration: The module keyword defines the boundary of the hardware block. The port list specifies the interface: `clk` is the system clock input that drives all sequential logic; `rst` is the asynchronous reset that immediately overrides normal operation; `main_st` and `cross_st` are 3-bit output registers that directly drive the traffic lights; and `light_count` is a 5-bit output register that exposes the internal timing counter.

State Parameter Encoding: The parameter declarations assign 2-bit binary values to the four FSM state names: `g_to_r` (2'b00), `y_to_r` (2'b01), `r_to_g` (2'b10), and `r_to_y` (2'b11). Using named parameters rather than raw binary values makes the code self-documenting and highly readable. These constants are synthesized away by the compiler and carry no hardware footprint.

Initial Block: The statement `initial light_count = 0` sets the power-on initial value of the counter to zero when simulation begins. On real FPGA hardware, this corresponds to the power-on reset value of the register.

Counter Always Block (always @ posedge clk or posedge rst): This block implements the 5-bit timing counter. The asynchronous reset (posedge `rst`) takes priority — when `rst` is asserted, `light_count` is immediately set to zero using a blocking assignment (`=`), which ensures the reset takes effect in the same time step. On normal clock edges, the counter increments using a non-blocking assignment (`<=`). When it reaches 31, it wraps to 0, completing the 32-second cycle.

State Transition Always Block (always @ posedge clk or posedge rst): This block controls the FSM state register. On asynchronous reset, it immediately returns to `g_to_r`. On each clock edge, the case statement evaluates the current state and conditionally transitions to the next state when the counter hits the programmed threshold. Critically, the counter is compared, not modified, here — separation of concerns makes the design cleaner and more maintainable.

Output Always Block (always @ posedge clk): This block assigns the traffic light output values based on the current FSM state. It is deliberately NOT sensitive to `rst` — the output will update

naturally on the next clock edge after reset, which is acceptable for this application. The 3-bit one-hot encoding (001 = green, 010 = yellow, 100 = red) is used to make decoding straightforward and to ensure only one light is active at a time per street.

Mixed Blocking and Non-Blocking Assignments: The design uses a mixture of blocking (=) and non-blocking (<=) assignments across different always blocks. Blocking assignments are used in the reset branches of counter and output logic for immediate effect; non-blocking assignments are used in the clocked branches to model proper register behavior and avoid race conditions between always blocks.

5.3 Verification Testbench Code

The testbench instantiates the traffic_controller module and applies clock and reset stimuli to verify correct behavior across multiple full 32-second cycles, including a mid-cycle reset scenario.

```
`timescale 1ns / 1ps
module traffic_controller_tb;
    reg clk;
    reg rst;
    wire [2:0] main_st;
    wire [2:0] cross_st;
    wire [4:0] light_count;

    traffic_controller uut (
        .clk(clk), .rst(rst),
        .main_st(main_st), .cross_st(cross_st), .light_count(light_count)
    );

    always begin
        #5 clk = ~clk;
    end

    initial begin
        $monitor("Time=%0dt rst=%b light_count=%0d | main_st=%b cross_st=%b",
            $time, rst, light_count, main_st, cross_st);
    end

    initial begin
        clk = 0; rst = 0;
        #10 rst = 1;
        #10 rst = 0;
        #700;
        #20 rst = 1;
        #10 rst = 0;
        #100;
        $finish;
    end
endmodule
```

5.4 Testbench Explanation

The testbench declares reg types for the input signals (clk and rst) and wire types for the output signals (main_st, cross_st, light_count), which is the standard pattern for Verilog testbenches. The Unit Under Test (UUT) is instantiated with named port connections.

Clock Generation: The always block toggles the clk signal every 5 time units (ns), creating a 10ns clock period. In the context of this simulation, each clock period represents 1 second of real-world time, making the 32-clock-cycle sequence correspond to the 32-second traffic light cycle.

Monitor Statement: The \$monitor system task continuously watches all signals and prints a formatted line to the simulation transcript whenever any signal changes. This provides a complete record of every state transition and counter value throughout the simulation without requiring manual inspection of waveforms.

Stimulus Sequence: The initial block sets clk and rst to 0, then asserts rst for 10 time units to reset the FSM and counter, then deasserts rst to begin normal operation. The simulation runs for 700 time units (approximately 70 clock cycles, enough for two complete 32-second cycles). A mid-cycle reset is then applied to verify that the FSM immediately returns to the initial state regardless of current position in the cycle.

Expected Behavior: After reset, the FSM enters g_to_r. light_count increments from 0. At count=15, the state transitions to y_to_r. At count=18, it moves to r_to_g. At count=28, it moves to r_to_y. At count=31, it wraps back to g_to_r and the cycle repeats. The mid-cycle reset test confirms that the controller recovers correctly and restarts from g_to_r with light_count=0.

6. RTL Schematic Layout

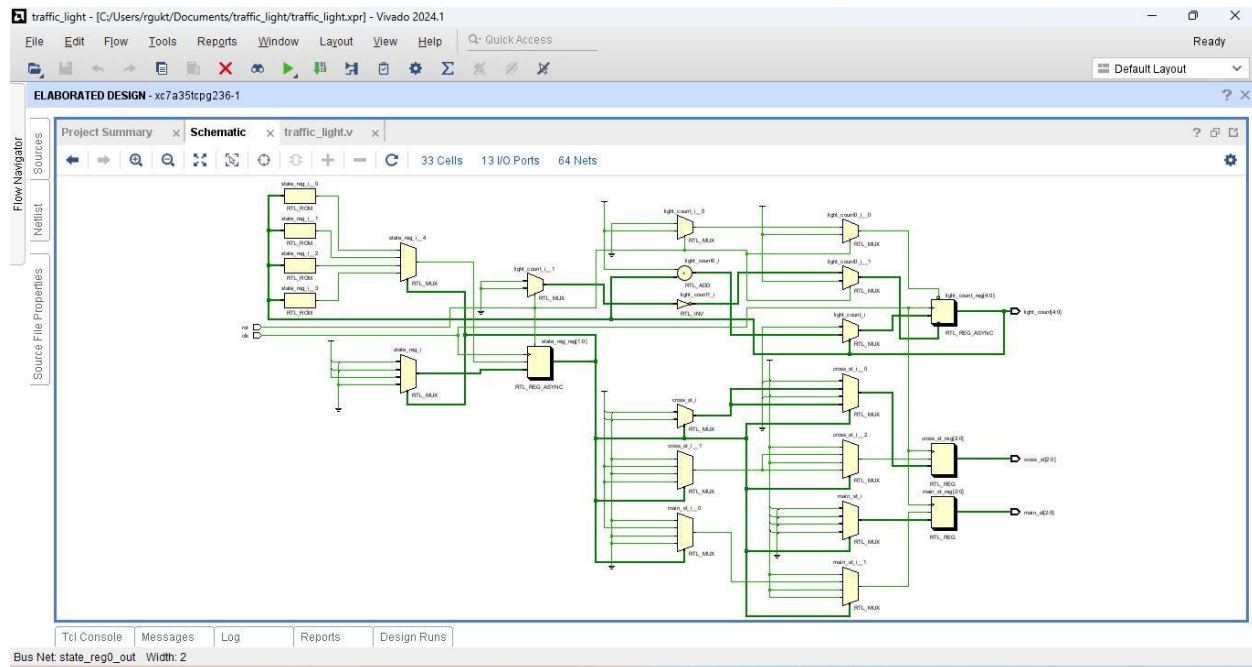
After synthesizing the Verilog design using an EDA tool such as Xilinx Vivado, the tool generates an RTL (Register Transfer Level) schematic. This schematic is a graphical representation of the synthesized hardware circuit that implements the Traffic Light Controller functionality.

The RTL schematic shows how the Verilog code has been translated into a network of logic gates, registers, counters, and multiplexers. Key elements visible in the RTL schematic include:

- **State Register (Flip-Flop Array):** The 2-bit state_reg is synthesized as two D flip-flops with synchronous enable and asynchronous reset. The flip-flops are visible as dedicated register elements in the schematic, fed by the next-state logic.
- **Counter Logic:** The 5-bit light_count register is synthesized as a chain of D flip-flops with an adder for increment logic and comparison circuits. The comparator outputs (count==15, count==18, count==28, count==31) are visible as logic gates feeding the state transition enable signals.

- **Next-State Logic:** Combinational logic gates (AND, OR) implement the state transition conditions. Each transition condition (e.g., `state==g_to_r` AND `count==15`) is synthesized into a product-of-sums or sum-of-products gate structure that feeds the D input of the state register flip-flops.
- **Output Decoder:** The case statement in the output always block is synthesized into a 2-to-3 decoder (with 4 valid input encodings) implemented using multiplexers or LUT (Look-Up Table) structures. Separate decoders drive `main_st` and `cross_st`.
- **Reset Network:** The asynchronous reset signal is connected directly to the asynchronous reset (CLR) pins of all flip-flops in the design, bypassing the clock and ensuring immediate reset without waiting for a clock edge.

The synthesized design maps efficiently to FPGA flip-flops and look-up tables (LUTs) without any combinational feedback loops. The presence of flip-flops in the schematic confirms that the design is correctly inferred as sequential (clocked) logic, consistent with the use of `always @(posedge clk)` blocks in the source code.



7. Simulation Results and Functional Output

7.1 Overview of Simulation

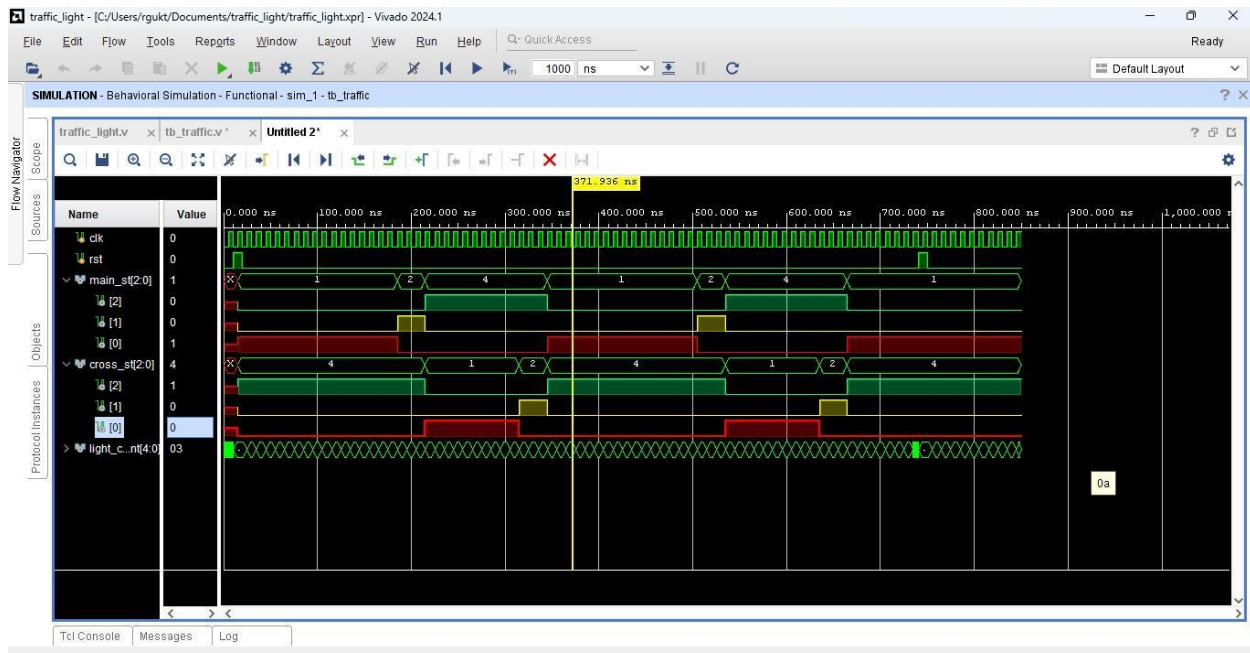
Simulation is the process of running the Verilog design and testbench in a software environment to verify that the hardware behaves as intended before physical implementation. Two industry-standard simulation tools are used in this project: Xilinx Vivado and Mentor ModelSim. Both tools read the Verilog source files, execute the testbench, and generate waveform outputs that can be inspected visually.

The simulation waveforms show the values of all signals (clk, rst, light_count, main_st, cross_st) plotted over time. Each state transition (triggered by light_count reaching its threshold) causes main_st and cross_st to update to new values on the next clock edge. By inspecting the waveform, the engineer can confirm that every state produces the correct light outputs and that the timing durations match the specification.

7.2 Vivado Simulation

Xilinx Vivado is a comprehensive FPGA design suite that includes a built-in behavioral simulator (XSim). To run simulation in Vivado, the design and testbench files are added to a project, and the "Run Simulation" option is selected from the Flow Navigator. Vivado compiles the Verilog files, elaborates the design hierarchy, and opens the waveform window showing the testbench signals over time.

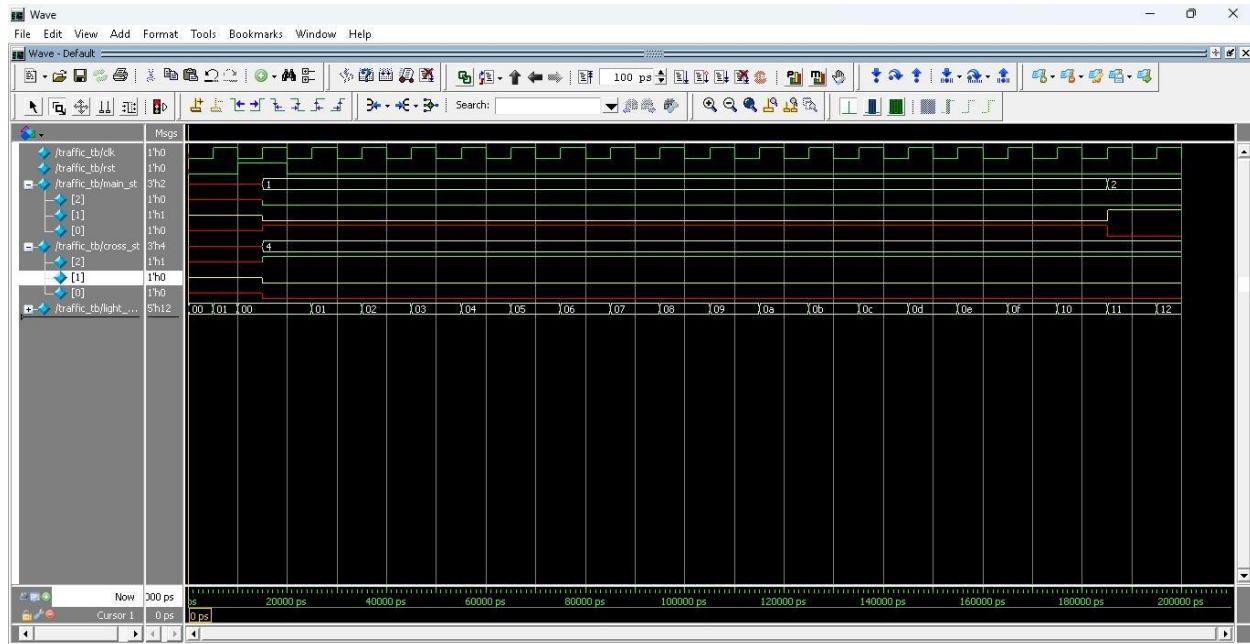
In the Vivado waveform window, each signal is displayed as a horizontal band. The clk signal appears as a regular square wave. The rst signal shows the brief high pulse at the beginning and later the mid-cycle reset pulse. The light_count signal is visible as a binary bus that increments from 0 to 31 then resets. The main_st and cross_st buses display the 3-bit one-hot values at each phase — the user can verify the correct transitions between 001 (green), 010 (yellow), and 100 (red).



7.3 ModelSim Simulation

Mentor ModelSim (also available as QuestaSim) is a standalone HDL simulation environment widely used in academia and industry. It supports both Verilog and VHDL simulation with a powerful waveform viewer. To simulate in ModelSim, the design and testbench files are compiled using the vlog command, the design is loaded with vsim, and waveforms are added to the Wave window using the add wave command. The simulation is then run for a sufficient number of time steps.

ModelSim provides advanced features such as signal grouping, radix formatting, and cursor-based time measurement, which make it easy to verify that each traffic light phase lasts for exactly the correct number of clock cycles. The \$monitor statement in the testbench also prints a complete text log of all signal changes to the ModelSim transcript window, providing a secondary verification channel. The \$finish statement automatically terminates the simulation after all test cases have run.



7.4 Simulation Analysis and Observations

Behavioral simulation confirms correct operation of the Traffic Light Controller under all conditions. The following observations were made from the simulation waveforms:

- **State Sequencing:** The FSM correctly cycles through all four states in order — $g_to_r \rightarrow y_to_r \rightarrow r_to_g \rightarrow r_to_y \rightarrow g_to_r$ — with each transition occurring at the precise counter value specified in the design.
- **Phase Durations:** Main green phase lasts exactly 15 clock cycles (counts 0–14); main yellow lasts 3 cycles (counts 15–17); cross green lasts 10 cycles (counts 18–27); cross yellow lasts 3 cycles (counts 28–30). Total cycle = 31 counts + wrap = 32 clock periods, matching the design specification.
- **Output Correctness:** main_st and cross_st show the correct one-hot encoded values in every state: in g_to_r , main_st = 001 and cross_st = 100; in y_to_r , main_st = 010 and cross_st = 100; in r_to_g , main_st = 100 and cross_st = 001; in r_to_y , main_st = 100 and cross_st = 010.
- **Safety Compliance:** At no point in the simulation do both main_st and cross_st simultaneously show a green output (3'b001), confirming that the mutual exclusion safety requirement is maintained throughout all transitions.
- **Reset Behavior:** When rst is asserted mid-cycle, the counter immediately clears to 0 and the FSM immediately returns to g_to_r on the same clock edge, demonstrating correct asynchronous reset behavior. After de-assertion, normal operation resumes from the beginning of the cycle.

- No Undefined States: No undefined (X) or high-impedance (Z) values appear on the output signals during any valid operating condition, confirming completeness of the case statement coverage and proper register initialization.
- Cycle Repetition: The simulation runs for more than two full 32-second cycles and both repetitions produce identical waveform patterns, confirming the design operates correctly as a continuous loop.

8. Future Scope

This Traffic Light Controller design provides a strong foundation that can be extended in several directions for academic research or industrial application:

1. Pedestrian Crossing Integration: A pedestrian request button input can be added to the controller. When pressed, the FSM can interrupt the normal cycle to insert a dedicated pedestrian crossing phase, adding a new state where all vehicle lights show red while pedestrian green is active.
2. Emergency Vehicle Override: A high-priority emergency override input can be wired to immediately set all lights to red (or the appropriate safe state) when an emergency vehicle is detected by a sensor. This is a critical safety feature in real-world traffic management systems.
3. Sensor-Based Adaptive Timing: The fixed timing parameters (15s, 3s, 10s, 3s) can be replaced by dynamic values driven by vehicle sensors or traffic density counters. An FPGA or microcontroller can read sensor data and adjust the counter thresholds in real time to optimize traffic flow.
4. FPGA Hardware Deployment: The synthesizable Verilog code can be downloaded onto a physical FPGA board (such as a Xilinx Basys 3 or Nexys A7) using Vivado's programming tools. LED outputs can be mapped to colored LEDs or a custom traffic light display connected to the FPGA's I/O pins.
5. Multiple Intersection Management: The single-intersection controller can be replicated and networked into a multi-intersection traffic management system. A central controller or distributed protocol (such as a green wave system) can synchronize multiple traffic light controllers to minimize vehicle stopping at successive intersections.
6. Fault Detection and Diagnostics: Built-in self-test (BIST) logic can be added to detect hardware faults such as stuck-at failures in the output flip-flops. If an illegal state (e.g., both streets green) is detected, the controller can enter a safe flashing-red mode and assert a fault output signal.

9. Conclusion

This project successfully demonstrates the design, implementation, and verification of a four-state Traffic Light Controller using Verilog HDL. The controller was designed as a finite state machine (FSM) using three clearly separated always blocks — one for timing counter logic, one for state transition logic, and one for output logic — following best practices in Verilog sequential design.

The design correctly implements a 32-second traffic signal cycle with main green (15s), main yellow (3s), cross green (10s), and cross yellow (3s) phases. The use of a 5-bit free-running counter as the timing reference and a 2-bit state register as the FSM element results in a clean, synthesizable, and easily maintainable design. The 3-bit one-hot output encoding ensures that only one light per street is active at any time, and the mutual exclusion of green signals across both streets is guaranteed by the FSM structure.

The Verilog source code is clean, synthesizable, and well-structured using named parameters for state encoding and clear comments for readability. The testbench systematically exercises the complete operating cycle and verifies correctness through simulation across multiple repetitions, including asynchronous reset behavior. Both Vivado and ModelSim simulations confirm that the design produces the expected outputs for all states with no undefined behavior.

The RTL schematic generated after synthesis reveals how the behavioral Verilog code translates into real hardware elements — flip-flops, counters, comparators, multiplexers, and reset networks — all correctly inferred by the synthesis tool. The design maps cleanly to FPGA resources with proper sequential logic structures and no unintended combinational feedback paths.

This project provides a complete, practical demonstration of sequential digital system design methodology: from behavioral FSM specification through HDL coding, simulation verification, and RTL synthesis. The knowledge and skills developed through this project form the basis for more advanced digital design work, including full processor pipeline implementation, embedded system control units, and custom ASIC development.

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